1 OCR Rockfest is a yearly music festival. Visitors buy tickets to attend and watch the music acts.

A computer system is created to sell tickets and store information about visitors and the music acts.

The number of different music acts on each of the three stages is stored in the array `actNumbers`.

Each index of the array represents one stage.

| Index | 0 | 1 | 2 |
|---|---|---|---|
| Data | 10 | 14 | 6 |

actNumbers

This algorithm adds up the total number of acts on all three stages.

```
count = 0
for x = 0 to 2
count = count + actNumbers[x]
next x
print(count)
```

i. Complete the trace table for the algorithm. You may not need to use all rows in the table.

| Line number | count | x | Output |
|---|---|---|---|
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |
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| | | | |
|---|---|---|---|
| | | | |

[4]

ii. The array `actNumbers` is changed to include data for two more stages. The new array is shown:

| Index | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| Data | 10 | 14 | 6 | 9 | 11 |

actNumbers

The algorithm from part **(i)** needs refining to work for the new array. The algorithm is repeated here:

```
01 count = 0

02 for x = 0 to 2

03 count = count + actNumbers[x]

04 next x

05 print(count)
```

Identify the line number that will need to be changed for the new array and write the updated line of code.

Line number _____

Updated code _____

_____ [2]

iii. Refine line 05 in the algorithm to output "`The total act count is `" and then the content of the variable `count`

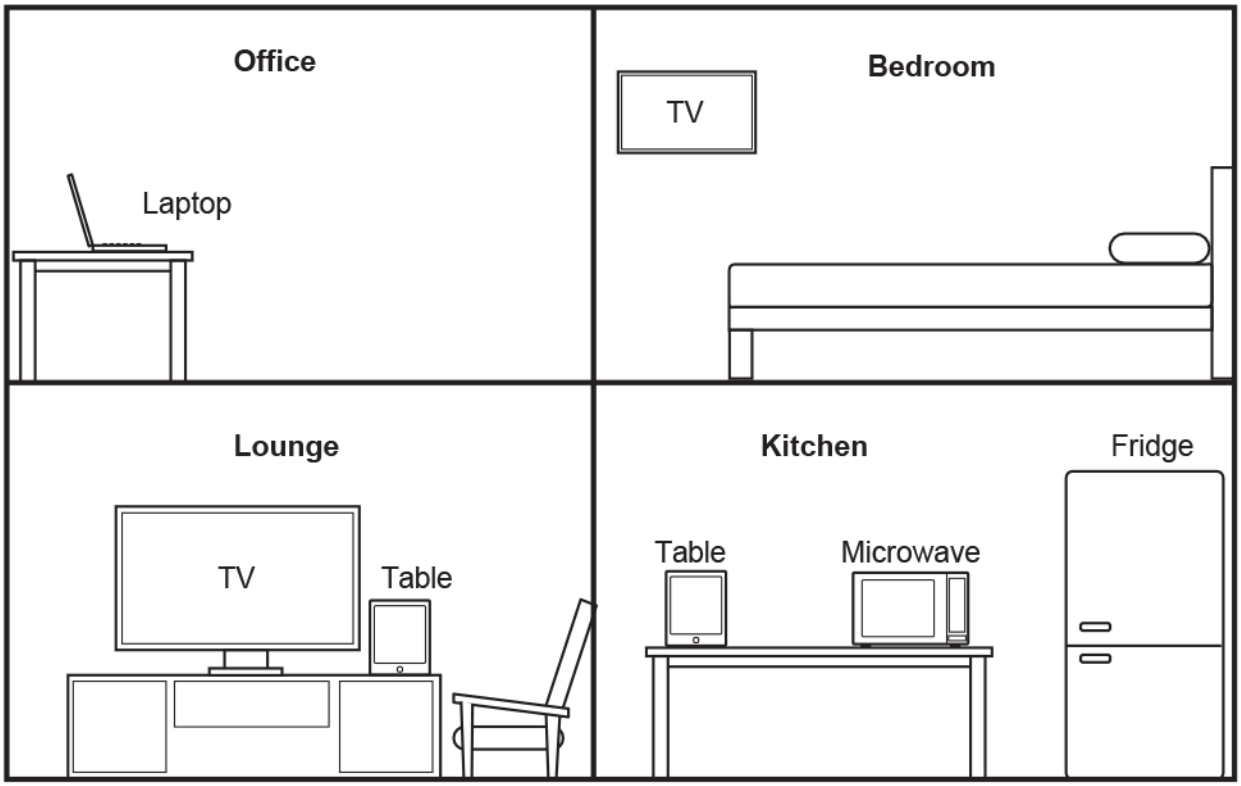
For example:

```
The total act count is 50
```

New line 05 _____

_____ [1]

2 Hope has a network in her house. The main devices are shown in the diagram.




Some of Hope's files are stored on the cloud.

Describe the benefits and drawbacks to Hope of storing her files on the cloud.

Benefits

Drawbacks

[6]

3 Eve's computer has system software including an Operating System and Utility Software.

Eve stores her computer program on the cloud whilst working on it.

i. Describe the benefits to Eve of storing the program on the cloud.

[3]

ii. Describe the drawbacks to Eve of storing the program on the cloud.

[3]

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4(a) A law company currently use a Local Area Network (LAN) linked to a Wide Area Network (WAN). They want to upgrade their system to utilise cloud storage.

Explain **two** advantages to the law company of storing their data in the Cloud.

1 _____

_____

_____

_____

2 _____

_____

_____

_____

**[4]**

(b) Explain **two** disadvantages to the law company of storing their data in the Cloud.

1 _____

_____

_____

_____

2 _____

_____

_____

_____

**[4]**

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5 Computer 1 enters the URL www.ocr.org.uk into a web browser. This is then converted into the IP address of the webserver that hosts the website.

Explain how the URL www.ocr.org.uk is converted into the IP address.

---

---

---

---

---

---

[3]

6 A house has computers in each room and a central router. Every room allows both Ethernet and WiFi connections to the router.

A user enters a uniform resource locator (URL) into a web browser on one of the computers in the house. A system is then used to find the IP address of the web server associated with the URL.

i. Name the system which matches URLs to IP addresses on the web.

---

---

[1]

ii. The following statements describe what happens after the IP address has been found and returned to the user's computer.

There are **five** missing statements in the table.

Write the letter of the missing statements from the table in the correct place to complete the description.

1 The request is put into packets

2 ----------

3 The packets are sent across the network

4 ----------

5 _____

6 If they have not arrived:

7 A timeout is sent to request the packets are resent

8 If they have arrived:

9 _____

10 _____

| Letter | Statement |
|---|---|
| A | The server checks if all the packets have arrived |
| B | The packets are put in order |
| C | The request is processed by the web server |
| D | The packets are received by the host server |
| E | Each packet is given the address and a number |

**[5]**

7 The artist uploads images to be displayed on a website. This is a client-server relationship.

i. Identify the computer that is acting as the client in this scenario **and** justify your choice.

Client computer _____

Justification _____

_____

_____

_____ [3]

ii. Identify the computer that is acting as the server in this scenario **and** justify your choice

Server computer _____

Justification _____

_____

_____

_____ [3]

END OF QUESTION PAPER

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<table>
<tr><th colspan="3">Question</th><th>Answer/Indicative content</th><th>Marks</th><th>Guidance</th></tr>
<tr><td>1</td><td></td><td>i</td><td>

- `Count` **initially** set to 0
- `count` increases to 10, 24, 30 and no other changes
- `x` goes from 0, 1, 2 (, 3) and no other changes.
- 30 output and no other output.

</td><td>4</td><td>

Example

<table>
<tr><th>Line number</th><th>count</th><th>x</th><th>Output</th></tr>
<tr><td>1</td><td>0</td><td></td><td></td></tr>
<tr><td>2</td><td></td><td>0</td><td></td></tr>
<tr><td>3</td><td>10</td><td></td><td></td></tr>
<tr><td>4</td><td></td><td></td><td></td></tr>
<tr><td>2</td><td></td><td>1</td><td></td></tr>
<tr><td>3</td><td>24</td><td></td><td></td></tr>
<tr><td>4</td><td></td><td></td><td></td></tr>
<tr><td>2</td><td></td><td>2</td><td></td></tr>
<tr><td>3</td><td>30</td><td></td><td></td></tr>
<tr><td>4</td><td></td><td></td><td></td></tr>
<tr><td>5</td><td></td><td></td><td>30</td></tr>
</table>

Ignore line numbers - not required. Mark order of values.

Values can be repeated in `count` and `x` columns, look at changes.

FT for BP4 if incorrect answer calculated but output. Allow **last** value of count calculated to be output for the mark.

Give BOD for “-”, “nothing”, etc as extra in output column. Do not accept `print(30)` or equivalent for output.

<u>Examiner’s Comments</u>

Trace tables are commonplace in Paper 2 and have examined before. Candidates showed better understanding of tracing a program through line by line, and this confidence was clear in candidate responses, with many able to correctly do this.

This question differed slightly from historical examination questions in two ways.

Firstly, this is the first time that a for loop has been asked about in a trace table. The challenge of the counter variable being updated was handled with ease by most candidates.

Secondly, although line numbers were asked for, these were not used in the marking of candidates’ responses and instead one mark was given per set of changes to variables. This process worked extremely well and will be used in the marking of any future trace table questions.

</td></tr>
</table>

| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| | | ii | • Line 02<br>• `For x = 0 to 4` | 2 | Must use `x` as counter variable.<br><br>Accept rewrites in other languages<br><br>BOD `0 to 5 / range(5)` etc<br><br>Accept other changes that implement the change successfully e.g. using length of `actNumbers`<br><br><u>Examiner's Comments</u><br><br>Candidates were able to confidently identify line 02 as the line to be changed and also identified the upper bound of the for loop as the value to be changed. Where candidates provided other solutions that logically worked, these were of course accepted. |

| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| | | iii | • `print("The total act count is " + count)` | 1 | Allow other concatenation operators e.g. & (VB), dot (PHP, Rust), double dot (Lua), etc. Allow comma. Must have some form of separator.<br><br>Ignore spacing. Ignore case. Ignore casting.<br><br>Allow other commands that are equivalent to `print` (e.g. `echo, output`, etc)<br><br>Allow multiple print statements. Allow multiple lines of code that achieve the outcome.<br><br>Allow string interpolation or language specific examples that will achieve this (e.g. f strings in Python, use of $ before variables inside strings in PHP). Both of the following and examples like this are OK :<br><br>• `print("The total act count is {count}")`<br>• `print("The total act count is $count")`<br><br><u>Examiner's Comments</u><br><br>As with many other smaller programming questions on this paper, candidates performed exceptionally well. The majority of candidates were able to use the **`count`** variable to output the correct value. Where this was not successfully done, candidates tended to include the variable identifier inside the string delimiters, thereby outputting the word count and not the value held in the variable.<br><br>Examiners were instructed to credit responses that used string interpolation, such as f strings in Python or the use of $ before strings in PHP. |
| | | | Total | 7 | |

| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| 2 | | | 1 mark per benefit to max 4, 1 mark per drawback to max 4<br>e.g.<br><br>Benefit<br><br>• Can access files from any device<br>• … e.g. they can instantly access the data from laptop and mobile phone<br>• Can access files from anywhere // Can access from anywhere with access to the Internet<br>• Can pay for auto-backups // don't have to backup manually<br>• Security **may be higher** than at home<br>• May be free of cost<br>• … you do not need to buy more hardware<br>• **Easier/quicker** to share files with others<br>• More available storage space on her device<br>• Easier to increase storage capacity (not memory)<br>• Can be used a backup in case of data loss<br><br>Drawback<br><br>• Cannot access files if no Internet access<br>• Not in control of security (bod less secure)<br>• … data may be hacked/stolen<br>• May cost monthly fee<br>• … which could be more long term than buying hardware<br>• May not be a backup // if cloud storage fails you have lost your data<br>• Data must be transferred to computer to read<br>• …. may be intercepted<br>• Connection may be slow<br>• … therefore takes time to upload/download<br>• May be issues as to who owns the data<br>• If stop paying / leave subscription other storage for files needs to be found<br>• If login details are forgotten/lost may not have access to files | 6 | Mark breadth and depth of knowledge.<br>1 mark for each valid point/expansion.<br>Allow specific examples as expansions for each point.<br>Mark benefits to max 4 first, then look for max 4 from drawbacks |
| | | | Total | 6 | |

| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| 3 | | i | 1 mark per bullet to max 3<br>e.g.<br><br>• She can access the program from anywhere<br>• … does not need to carry a storage device with her<br>• Security/backup is (likely) managed for her<br>• …does not need to manually backup his work | 3 | |
| | | ii | 1 mark per bullet to max 3<br><br>• If there is no Internet access he cannot access his work<br>• Transmission may not be secure<br>• …his work could be intercepted<br>• Security is out of his control<br>• …it may not be backed up/kept safe | 3 | |
| | | | Total | 6 | |

| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| 4 | a | | Two advantages from:<br><br>• It would offer additional storage (1) so the company can take on more cases (1)<br>• It is a very efficient method of backing up data (1) and so saves the firm time and money (1)<br>• It would allow their employees to work from anywhere (1) so they can take cases from other countries (1)<br>• It is environmentally friendly (1)<br>• Easy to increase availability of storage (1)<br>• You don't need specialist network skills (1) so the firm don't need to employ more staff (1)<br>• The third party provides security (1) so the company saves money on staff and software/hardware (1)<br>• The third party provides backup (1) so the company saves money on staff and software/hardware (1)<br>• Cheaper as don't need own infrastructure (1)<br><br>Each advantage needs to be contextualised to gain 2 marks. | 4<br>(AO2 1b) | 1 mark to be awarded for each correct advantage, with a mark for a discussion of the advantage related to the law firm. To a maximum of 2 advantages.<br><br>The total number of marks to be awarded for this task is 4 marks.<br><br>Responses which are not contextualised will gain a maximum of 1 mark per advantage (to a maximum of 2 advantages). |
| | b | | Two disadvantages from:<br><br>• You need a constant internet connection (1) which lawyers who travel a lot may not always have (1)<br>• Reliant on third party to carry out security procedures (1) but the firm are still legally responsible if things go wrong (1)<br>• Reliant on third party for back up connection (1)<br>• Data stored in the Cloud will be vulnerable to hacking and other threats (1) which the firm have no control over (1)<br>• Issues regarding data ownership (1)<br>• Implications of Data Protection Act (1)<br><br>Each disadvantage needs to be contextualised to gain 2 marks | 4<br>(AO2 1b) | 1 mark to be awarded for each correct disadvantage with a mark for a discussion of the disadvantage related to the law firm. To a maximum of 2 disadvantages.<br><br>The total number of marks to be awarded for this task is 4 marks.<br><br>Responses which are not contextualised will gain a maximum of 1 mark per disadvantage (to a maximum of 2 disadvantages). |
| | | | Total | 8 | |

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| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| 5 | | i | 1 mark per bullet to max 3<br><br>• URL sent to DNS // request sent to DNS for/with URL<br>• DNS **looks up/finds** to IP in its database<br>• DNS returns IP<br>• IF not found, DNS sends to higher level DNS | 3 | Request sent to DNS is NE without saying the URL is sent.<br><br>Only penalise missing or incorrect term for DNS once then FT |
| | | ii | 1 mark per bullet to max 3<br>e.g.<br><br>• **Destination** IP/address<br>• **Sender** IP/address<br>• Packet Number<br>• Packet size<br>• Number of packets<br>• Error detection method/value | 3 | Do not award MAC address<br><br><u>Examiner's Comments</u><br>Question (c) required candidates to apply their understanding of virtual memory and convert this into an algorithm. Candidates tackled this question well; they considered the possible answers and many selected correct responses. The most challenging aspect was to check if NOT (data required is in RAM).<br><br>Candidates were given the opportunity to argue whether a tablet is an embedded system, or not. With either answer being acceptable but candidates were required to justify their choice. This required application of the question to the scenario of the tablet. Candidates often gave generic definitions of what an embedded system, without any application to the tablet and whether this meets the definition of an embedded system or not. |
| | | | Total | 3 | |

| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| 6 | | i | Domain Name Server // DNS. | 1<br>AO1 1a<br>(1) | Allow Server/service/system<br><br><u>Examiner's Comments</u><br><br>This question required an understanding of a Domain Name Server/System to convert URLs to IPs. Few candidates were able to identify this system, with many making guessing such as HTTP. Many candidates did not attempt to answer this question. |
| | | ii | 1 mark for each letter in the correct place<br>1 The request is put into packets<br>2 E<br>3 The packets are sent across the network<br>4 D<br>5 A<br>6 If they have not arrived:<br>7 A timeout is sent to request the packets are resent<br>8 If they have arrived:<br>9 B<br>10 C | 5<br>AO1 1b<br>(5) | <u>Examiner's Comments</u><br><br>This question required an understanding of how packets of data are sent across a network. Candidates were required to read through the statements and order them logically. The more able were able to identify the appropriate sequence. Less able candidates confused some of the statements, such as identifying the request as being processed before the packets were received. |
| | | | Total | 6 | |

| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| 7 | | i | 1 mark for identification:<br><br>• Artist's computer // computer uploading the images // BOD The artist<br><br>1 mark each for justification to max 2:<br>e.g.<br><br>• Sends the files/data for storage/to the **host**/web **server** // the files are stored on the web **server**<br>• Performs the user's actions …<br>• .. and sends the results to the web **server**<br>• Sends a **request** to the web **server**…<br>• … to **store/upload** its files<br>• It does not store data for others to access<br>• Confirmation of upload/error is received (from server) for display | 3 | Incorrect computer, do not award justification.<br><br>Be careful the justification is talking about the upload of images to the web server, not the download.<br><br>Accept host for web server.<br><br>If 'user's computer' is given for identification, this is NE – read on for justification. If 'user viewing the website' or similar is given this is incorrect.<br><br><u>Examiner's Comments</u><br><br>Many responses identified the artist's computer as being the one that is acting as a client.<br><br>Fewer responses were able to justify this, for example they described a different scenario, such as downloading the images to view the website, instead of the given scenario of the client uploading the files.<br><br>Exemplar 2<br><br>Client computer: Artist's Computer<br>Justification: The Artist's Computer is requesting a response for uploading images by the website<br>[3]<br><br>In this response the candidate has identified the correct computer. The justification states that it is requesting a response, but not where this response is from, and then that the images are uploaded to the website when in this scenario they should be referring to the server in the client-server relationship. |

| Question | | | Answer/Indicative content | Marks | Guidance |
|---|---|---|---|---|---|
| | | ii | 1 mark for identification:<br><br>• Web server<br><br>1 mark each for justification to max 2:<br>e.g.<br><br>• The images/data are stored on / uploaded to / sent to / hosted on the web server<br>• Web server receives a request (from the artist's computer to upload the images)<br>• Web server executes/responds to the request // Web server is doing the processing/handling the (request to) upload<br>• Web server returns confirmation/error of the processing/upload | 3 | If computer is incomplete or inaccurate e.g. server/website instead of web server. Do not award computer, but award justification.<br>Allow FT in justification if the same inaccurate term is used, for example 'website' is given as computer (NE), but justification is: 'images are sent to the website' (FT for website instead of web server).<br><br>Incorrect computer, do not award justification.<br><br><u>Examiner's Comments</u><br><br>Fewer candidates were able to accurately identify the server in this scenario.<br><br>Many candidates identified the website as a server when a website is not a computer.<br><br>Some candidates who identified the webserver were also able to justify their choice by identifying the actions it was performing in the scenario.<br><br>Exemplar 3<br><br>Server computer website server<br>Justification the website server receives and processes data and does not send it back<br><br>The response in Exemplar 3 has correctly identified the webserver as the computer. They have also identified that this server (the computer) receives the data and processes the data. |
| | | | Total | 6 | |